

## Assignment-2 Session11

**Question 1:** Pick any one feature from an app you use daily (for example: "Search Restaurants" in Zomato or "Add to Cart" in Flipkart) and break down the testing tasks for this feature using a Work Breakdown Structure (WBS). List at least 6 distinct testing activities or sub-tasks.

**Answer**

**Application:** Flipkart

**Feature:** Add to Cart

**Work Breakdown Structure (WBS)**

| WBS No. | Testing Activity       | Description                                                                 |
|---------|------------------------|-----------------------------------------------------------------------------|
| 1.0     | Requirement Analysis   | Study the functional requirements of the Add to Cart feature.               |
| 2.0     | Test Planning          | Prepare the testing strategy, schedule, and required resources.             |
| 3.0     | Test Case Design       | Create positive, negative, and boundary test cases.                         |
| 4.0     | Test Environment Setup | Configure browsers, test accounts, and the testing environment.             |
| 5.0     | Test Execution         | Execute all test cases and compare actual and expected results.             |
| 6.0     | Defect Reporting       | Report defects, assign severity and priority, and track them until closure. |
| 7.0     | Regression Testing     | Verify that fixing defects has not affected existing functionality.         |
| 8.0     | Test Closure           | Prepare the Test Summary Report and complete testing activities.            |

**Question 2:** For the "Send Message" feature in WhatsApp, estimate the total test effort required by listing all the main test activities, assigning estimated hours to each, and calculating the overall effort needed.

**Answer**

| Activity               | Estimated Hours |
|------------------------|-----------------|
| Requirement Analysis   | 2 Hours         |
| Test Planning          | 2 Hours         |
| Test Case Design       | 5 Hours         |
| Test Environment Setup | 2 Hours         |

| Activity            | Estimated Hours |
|---------------------|-----------------|
| Test Execution      | 8 Hours         |
| Defect Reporting    | 3 Hours         |
| Retesting           | 2 Hours         |
| Regression Testing  | 4 Hours         |
| Test Summary Report | 2 Hours         |

**Total Estimated Effort**

**Total Hours = 30 Hours**

**Question 3:** Flipkart's "Apply Coupon" feature is being released as a small module. Create a simple test effort estimation sheet showing each test activity, estimated hours, and assumptions.

**Answer**

**Test Effort Estimation Sheet**

| Activity               | Estimated Hours | Assumptions                                         |
|------------------------|-----------------|-----------------------------------------------------|
| Requirement Analysis   | 2 Hours         | Requirements are complete and approved.             |
| Test Planning          | 2 Hours         | One tester is assigned.                             |
| Test Case Design       | 4 Hours         | Around 20 test cases will be created.               |
| Test Environment Setup | 2 Hours         | Test environment is available.                      |
| Test Execution         | 6 Hours         | No major environment issues occur.                  |
| Defect Reporting       | 2 Hours         | Few defects are expected.                           |
| Retesting              | 2 Hours         | Developers fix defects on time.                     |
| Regression Testing     | 3 Hours         | Existing checkout functionality needs verification. |
| Test Closure           | 1 Hour          | Test Summary Report will be prepared.               |

**Total Estimated Effort**

**Total Hours = 24 Hours**

**Question 4:** Read about Test Point Analysis (TPA) as introduced in class. Explain how TPA could help estimate testing effort for a new "Movie Booking" feature in BookMyShow. List two advantages and one limitation.

**Answer**

**Test Point Analysis (TPA)**

Test Point Analysis (TPA) is a software testing estimation technique used to estimate the testing effort based on the complexity, size, and functionality of the application. It helps testers estimate the amount of work required before testing begins.

### **Example**

For the **Movie Booking** feature in BookMyShow, TPA can be used to estimate the effort required to test functionalities such as movie search, theatre selection, seat booking, payment processing, booking confirmation, and ticket cancellation. More complex features receive higher test points and therefore require more testing effort.

### **Advantages of TPA**

#### **Provides More Accurate Estimates**

Testing effort is estimated based on application complexity instead of guesswork.

#### **Improves Resource Planning**

Helps allocate testers, schedule activities, and estimate project timelines effectively.

### **Limitation of TPA**

#### **Requires Well-Defined Requirements**

If requirements are incomplete or frequently changing, the estimation may become inaccurate.